MULTI-SCALE LARGE KERNEL ATTENTION FOR SINGLE-IMAGE DERAINING

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ABSTRACT

Rain streaks inherently exhibit spatial heterogeneity. Although recent hybrid CNN–Transformer single-image deraining(SID) methods benefit from complementary local and global modeling to recover clear images from the degraded ones. However, these methods consistently suffer from module redundancy and suboptimal feature fusion, due to the co-designed CNN–Transformer architecture and the resulting misalignment in learned representations. In this paper, we propose a Multi-Scale Large-Kernel Attention (MLA) module and a Ghost Kernel Gate (GKG) to jointly model long-range dependencies and local details within a unified convolutional space. Specifically, MLA captures rich multi-scale contextual features, while GKG enables local features selection. The unified-space multi-scale feature extraction and fusion strategy effectively handles spatially-varying rain distribution and reduces model parameters. Extensive experiments demonstrate that our method achieves state-of-the-art performance, yielding consistent improvements in both quantitative metrics and visual quality.

***Index Terms*—** Large-kernel, Multi-scale, Self-attention, Single-image deraining

1. INTRODUCTION

Images captured in rainy conditions often suffer from severe degradation, challenging downstream vision tasks like object tracking [1] and autonomous driving [2]. Single-image deraining (SID) aims to recover clear images, but the multi-scale features and spatial heterogeneity of rain streaks pose significant challenges. CNN-based methods excel at local feature extraction but struggle with large-scale rain removal and global consistency [3, 4]. Transformer-based approaches model long-range dependencies effectively, but often incur high computational cost and compromise fine-grained detail preservation [5, 6]. Hybrid CNN-Transformer models seek a balance but often face redundancy and misalignment between feature spaces, limiting performance [7] [8].

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Recent advances in large-kernel network design have validated large-kernel convolutions as an effective mechanism for capturing global context. RepLKNet [1] expands the receptive field with large kernels to model long-range dependencies, LSKA [9] employs large separable kernels to improve efficiency while preserving contextual modeling, and VAN [10] leverages large-kernel attention to enhance feature representations. Motivated by these advances and the spatial heterogeneity of rain-streak patterns, we propose a Multi-Scale Large-Kernel Attention Deraining Network (MSLKAD), integrating (i) a Multi-Scale Large-Kernel Attention (MLA) module for multi-scale, fine-grained feature extraction, (ii) a Ghost Kernel Gate (GKG) for adaptive local feature refinement, and (iii) an Expanded Large-Kernel Attention (EKA) module to enlarge the receptive field and enforce global consistency. This unified multi-scale extraction and fusion strategy effectively handles spatially varying rain streaks while reducing model complexity.

The main contributions are summarized as follows:

- We propose a multi-scale large-kernel attention mechanism for image deraining, which effectively captures the spatial variations and intensity changes of rain streaks.
- We develop an MLA module for multi-scale large-kernel contextual aggregation, and further introduce GKG and EKA to adaptively refine local details and expand the receptive field, jointly improving detail fidelity and global consistency.
- Extensive experiments on both synthetic and real-world datasets demonstrate that the proposed method achieves state-of-the-art performance and strong robustness.

2. METHODOLOGY

2.1. Overall Framework

The overall architecture of our Multi-Scale Large Kernel Attention Deraining network (MSLKAD) is illustrated in Figure 1. Given a rainy input $I_{\text{rain}} \in \mathbb{R}^{C\times H\times W}$, MSLKAD first maps it into a high-dimensional feature space via a convolutional head:

$$F_0 = H_{\text{head}}(I_{\text{rain}}), \tag{1}$$

where $H_{\text{head}}(\cdot)$ is a 3×3 convolution. The network body consists of N_0 stacked Multi-Feature Attention Blocks (MFABs),

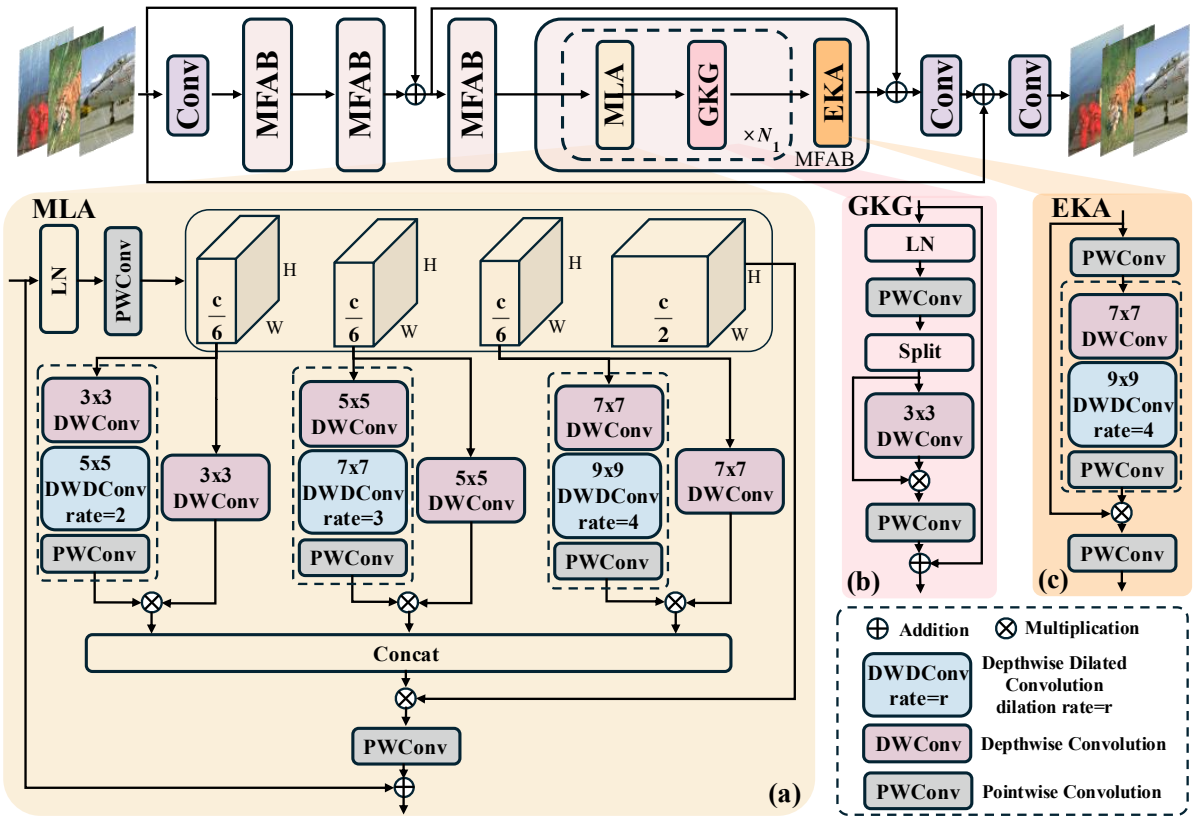


Fig. 1: Illustration of the proposed MSLKAD framework. Given a rainy input image, the network generates a high-quality derained output. The architecture consists of (a) MLA for joint long-range and local feature modeling, (b) GKG for rain suppression and detail enhancement, and (c) EKA for enlarging the receptive field to handle complex, non-uniform rain streaks.

each containing N_1 pairs of Multi-Scale Large Kernel Attention (MLA) and Ghost Kernel Gate (GKG) modules:

$$F_{i,j}^{\text{out}} = \text{GKG}(\text{MLA}(F_{i,j}^{\text{in}})), \quad i \in [1, N_0],\ j \in [1, N_1] \tag{2}$$

with the MFAB output globally aggregated by an Expanded Large Kernel Attention (EKA) module:

$$F_i^{\text{out}} = \text{EKA}(F_{i,N_1}^{\text{out}}). \tag{3}$$

After progressive feature enhancement across all MFABs, the final feature is fused with the head feature F_0 and mapped back to image space via a convolutional tail:

$$I_{\text{derain}} = H_{\text{tail}}(F_{N_0}^{\text{out}} + F_0), \tag{4}$$

where $H_{\text{tail}}(\cdot)$ is a 3×3 convolution.

We employ pixel-level, perceptual, and edge constraints to supervise training:

$$\mathcal{L}_{\text{L1}} = \|I_{\text{derain}} - I_{\text{gt}}\|_1, \tag{5}$$

$$\mathcal{L}_{\text{PL}} = \|\phi(I_{\text{derain}}) - \phi(I_{\text{gt}})\|_1, \tag{6}$$

$$\mathcal{L}_{\text{EDGE}} = \|\nabla I_{\text{derain}} - \nabla I_{\text{gt}}\|_1, \tag{7}$$

with the total loss,

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{L1}} + \lambda_2 \mathcal{L}_{\text{PL}} + \lambda_3 \mathcal{L}_{\text{EDGE}}, \tag{8}$$

where $\phi(\cdot)$ is the VGG19 feature extractor, and $\{\lambda_1, \lambda_2, \lambda_3\} = \{1, 0.7, 0.15\}$.

2.2. Multiscale Large Kernel Attention (MLA)

Within each MFAB, the MLA module captures multi-scale contextual information and enhances local details, which is crucial for removing non-uniform rain streaks (see Figure 1(a)). To address the limitations of conventional networks in modeling long-range dependencies and fine-grained textures, MLA integrates large kernel attention, multi-scale decomposition, and residual recalibration.

Large Kernel Attention. Given $F_{\text{in}} \in \mathbb{R}^{C \times H \times W}$, LKA decomposes a large $K_i \times K_i$ convolution with dilation d_i into depth-wise, dilated, and point-wise convolutions:

$$\text{LKA}_{\{K_i, d_i\}}(F_{\text{in}}) = f_{\text{PW}}\big(f_{\text{D}}(f_{\text{DW}}(F_{\text{in}}))\big), \tag{9}$$

effectively capturing both global rain patterns and structural dependencies.

Table 1: Quantitative evaluations on synthetic and real datasets. **Bold** and underline indicate the best and second-best results.

Method	Synthetic				Real							
	Rain200L [11]		Rain200H [11]		SPA-Data [12]		RealRain-1k-L [13]		RealRain-1k-H [13]		LHP-Rain [14]	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
(ICCV'15) DSC [15]	27.16	0.8663	14.73	0.3815	34.95	0.9416	27.76	0.8750	24.48	0.7857	-	-
(CVPR'16) GMM [16]	28.66	0.8652	14.50	0.4164	34.30	0.9428	28.87	0.9259	24.93	0.8712	-	-
(CVPR'17) DDN [17]	34.68	0.9671	26.05	0.8059	36.16	0.9457	31.18	0.9172	29.17	0.8783	32.61	0.9037
(ECCV'18) RESCAN [18]	36.09	0.9697	26.75	0.8353	38.11	0.9707	31.33	0.9261	29.39	0.8914	32.72	0.9058
(CVPR'19) PReNet [19]	37.80	0.9814	29.04	0.8991	40.16	0.9816	30.40	0.9444	27.02	0.9040	33.17	0.9177
(CVPR'20) MSPFN [3]	38.58	0.9827	29.36	0.9034	43.43	0.9843	35.51	0.9668	25.08	0.8045	32.20	0.8955
(AAAI'21) DualGCN [20]	40.73	0.9886	31.15	0.9125	44.18	0.9902	34.17	0.9735	30.18	0.9500	33.38	0.9200
(ICCV'21) SPDNet [4]	40.50	0.9875	31.28	0.9207	43.20	0.9871	28.98	0.9005	26.63	0.8498	32.98	0.9122
(TPAMI'22) IDT [11]	40.74	0.9884	32.10	0.9344	47.35	0.9930	36.24	0.9789	33.97	0.9729	33.43	0.9246
(CVPR'22) Restormer [21]	40.99	0.9890	32.00	0.9329	47.98	0.9921	40.90	0.9849	39.57	0.9812	33.77	0.9290
(CVPR'23) DRSformer [6]	41.23	0.9894	32.17	0.9326	48.54	0.9924	40.93	0.9863	38.49	0.9786	33.55	0.9227
(ICCV'23) UDR-S2Former [22]	40.96	0.9892	32.59	0.9374	48.57	0.9917	42.07	0.9890	39.75	0.9847	34.74	0.9489
(AAAI'24) MSDT [5]	41.75	0.9904	32.45	0.9379	49.07	0.9926	42.01	0.9884	40.95	0.9860	34.53	0.9466
(CVPR'24) NeRD-Rain [23]	41.71	0.9903	32.40	0.9373	49.58	0.9940	42.54	0.9904	41.05	0.9858	34.78	0.9484
(Neural Netw.'25) PADUM [24]	**42.24**	**0.9917**	32.85	0.9402	49.76	0.9940	42.32	0.9891	41.10	0.9868	34.88	0.9493
(Ours) MSLKAD	42.11	0.9909	**33.09**	**0.9435**	**50.12**	**0.9949**	**42.93**	**0.9924**	**41.94**	**0.9896**	**35.62**	**0.9507**

Multi-Scale Mechanism. F_{in} is first projected to double the number of channels via a 1×1 convolution and split into an attention branch A and a residual branch X. A is further divided into three groups A_i corresponding to different scales, each processed by an LKA with kernel–dilation pairs $\{K_i, d_i\}$, where $\{7, 2\}$, $\{21, 3\}$, and $\{35, 4\}$ are adopted in cascaded form (3-5-1, 5-7-1, 7-9-1):

$$A'_i = \text{LKA}_{\{K_i,d_i\}}(A_i) \cdot G_i(A_i), \quad i = 1, 2, 3, \tag{10}$$

where $G_i(\cdot)$ is a gating convolution. The multi-scale outputs are concatenated, fused with X, scaled by a learnable parameter α, and added to F_{in}:

$$F_{\text{out}} = \alpha \cdot \left(X \cdot \text{concat}(A'_i)_{i=1}^{3} \right) + F_{\text{in}}. \tag{11}$$

This design enables small kernels to capture fine rain streaks while allowing large kernels to model long-range dependencies, integrating multi-scale attention with residual learning to enhance both global context and local details for fine-grained restoration of non-uniform rain.

2.3. Ghost Large Kernel Gating (GKG)

Inspired by gated attention units, GKG integrates simple spatial attention and gated linear units to selectively enhance meaningful textures while suppressing redundancy (Figure 1(b)). Given channel-expanded features F, they are split into X and B, with B passing through a depth-wise convolution to generate a spatial gate:

$$F_{\text{GKG}} = X \cdot f_{\text{DW}}(B) + F, \tag{12}$$

where $f_{\text{DW}}(\cdot)$ denotes depth-wise convolution. This enables efficient local feature enhancement with low computational cost.

2.4. Expanded Large Kernel Attention (EKA)

While MLA and GKG capture multi-scale features and long-range dependencies, some global structures may still be underrepresented. To address this issue, EKA is placed at the block tail (Figure 1(c)) and we employ a 7-9-1 LKA, composed of cascaded depth-wise, dilated, and pointwise convolutions, This helps generate spatial attention maps and refine features globally. This design complements MLA and GKG by enhancing cross-region dependencies, preserving fine textures, and improving overall deraining performance.

3. EXPERIMENTS

3.1. Datasets and Implementation Details

Datasets. We evaluate MSLKAD on a comprehensive suite of six benchmarks to ensure rigorous validation. This includes two synthetic datasets (Rain200L/H [11]) representing diverse rain streak densities, and four real-world datasets (SPA-Data [12], RealRain-1K-L/H [13], LHP-Rain) to assess generalization capabilities in complex, uncontrolled scenarios. Rain200L/H consist of 1800/200 (train/test) pairs each. For real-world evaluation, SPA-Data contains 638,492/1000 pairs, RealRain-1K-L/H provides 784/224 pairs, and LHP-Rain includes 2100/300 pairs.

Comparison Methods. We benchmark MSLKAD against a wide spectrum of representative approaches, including prior-based (DSC [15], GMM [16]), CNN-based (DDN [17], RESCAN [18], PReNet [19], DualGCN [20], SPDNet [4]), and Transformer-based methods (Restormer [21], IDT [11], DRSformer [6], UDR-S2Former [22], NeRD-Rain [23], MSDT [5]). Pretrained models are used when available; otherwise, models are retrained under standard settings. Additional results are cited from [13, 14].

Evaluation Metrics. We report Peak Signal-to-Noise Ra-



Fig. 2: Visual comparison results on synthetic datasets. The top row shows the results on the Rain100H [11] dataset, while the bottom row shows the results on the Rain100L [11] dataset.

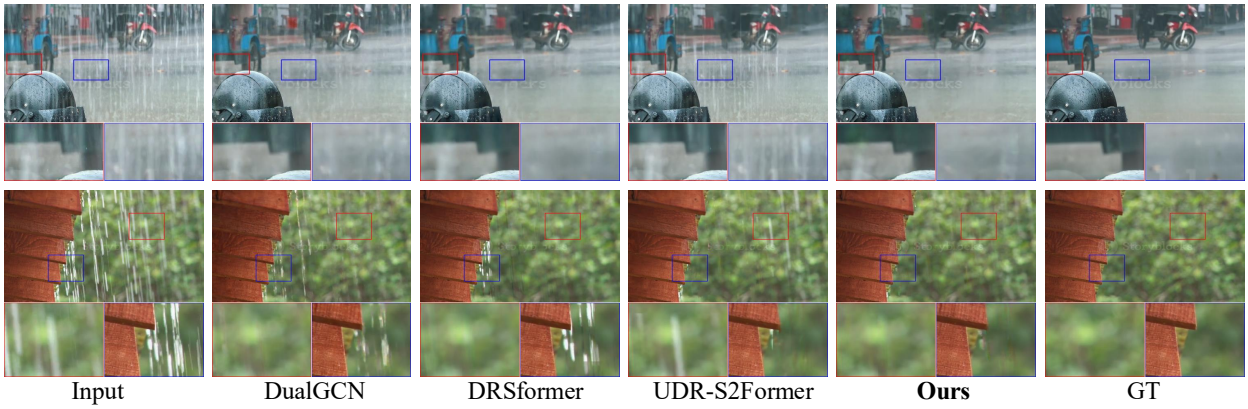


Fig. 3: Visual comparison results on real-world datasets. The top row shows the results on the RealRain-1k-H [13] dataset, while the bottom row displays the results on the RealRain-1k-L [13] dataset.

tio (PSNR) [25] and Structural Similarity Index (SSIM) [26]. Following standard conventions in deraining tasks, these metrics are calculated on the luminance (Y) channel of the YCbCr color space.

Training Details. MSLKAD is implemented in PyTorch with depth $(N_0, N_1) = (4, 9)$ and channels $C = 180$. We train on four NVIDIA RTX 3090 GPUs using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$) with batch size 8 and patch size 128×128. The learning rate decays from 10^{-4} to 10^{-6} via cosine annealing. SPA-Data is trained for 400K iterations, others for 100K. Standard data augmentations are employed.

3.2. Comparison with the State-of-the-Arts

Synthetic Dataset Results. As shown in Table 1, MSLKAD consistently outperforms prior art, notably surpassing NeRD-Rain [23] by 0.40 dB and 0.69 dB on Rain200L and Rain200H [11]. Qualitatively (Fig. 2), while CNN-based methods struggle with dense rain and Transformers like DRSformer often blur fine details, our MSLKAD effectively restores structural elements by balancing multi-scale modeling and local refinement. Beyond restoration quality, we further evaluate the computational efficiency; detailed comparisons of parameters and inference latency are provided in Table 4.

Real-world Dataset Results. On real-world datasets (Table 1, last four columns), MSLKAD generalizes across varying rain distributions. It surpasses 50 dB on SPA-Data [12]. On RealRain-1k [13], MSLKAD outperforms NeRD-Rain, obtaining 42.93 dB on its Light subset and 41.94 dB on the Heavy subset. Visual comparisons in Figure 3 corroborate these quantitative findings. In real-world scenarios, such as the street view (top row) and architectural textures (bottom row), competing methods like DualGCN and UDR-S2Former tend to suffer from over-smoothing or residual rain haze. Conversely, MSLKAD removes irregular rain streaks while preserving image textures and color fidelity, demonstrating

Method	X			Y			Z			Rain100H [11]	
	3-5-1	5-7-1	7-9-1	3-5-1	5-7-1	7-9-1	3-5-1	5-7-1	7-9-1	PSNR	SSIM
MLA	✓			✓			✓			32.62(↓ 0.47)	0.9371
		✓			✓			✓		32.84(↓ 0.25)	0.9394
			✓			✓			✓	32.88(↓ 0.21)	0.9401
	✓					✓				32.69(↓ 0.40)	0.9374
		✓				✓				32.74(↓ 0.35)	0.9382
	✓								✓	32.65(↓ 0.53)	0.9373
	✓				✓				✓	**33.09**	**0.9435**

Table 2: Ablation of MLA on Rain100H. Columns X, Y, Z denote three branches with kernel groups 3-5-1, 5-7-1, 7-9-1; tick marks indicate active kernels. Individual kernel details are omitted for clarity.

Table 3: Ablation of MSLKAD components on Rain100H, showing the effects of EKA, multi-scale mechanism, and GKG. EKA is replaced with a convolution, multi-scale with LKA (5-7-1), and GKG with an MLP.

Method	EKA	GKG	MLA	PSNR	SSIM
MSLKAD				30.04	0.9024
	✓			30.18 (↑ 0.14)	0.9066
	✓		✓	30.68 (↑ 0.64)	0.9123
	✓	✓		30.99 (↑ 0.81)	0.9173
	✓	✓	✓	33.09 (↑ 3.05)	0.9435

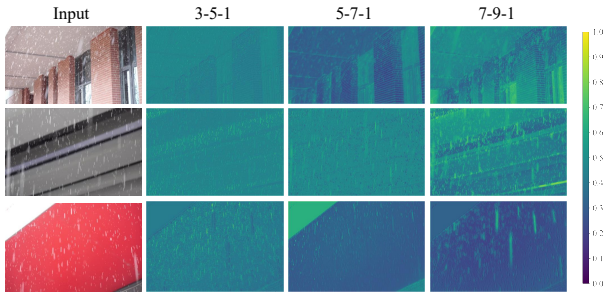


Fig. 4: Feature visualization of different branches in the MLA. The (7, 9, 1) branch captures rain streaks more effectively benefiting from the larger receptive field.

its capability in handling real-world degradations.

3.3. Ablation studies

We conduct a comprehensive ablation analysis under a standard 100K-iteration training setup to validate the efficacy of each proposed component, with quantitative results reported in Table 3. Starting from a baseline model, the integration of the Multi-scale Linear Attention (MLA) module yields a substantial improvement in PSNR, demonstrating its critical role in modeling long-range dependencies and global rain patterns. Furthermore, the addition of the Gated Kernel Attention (GKA) provides complementary benefits by refining local features and suppressing high-frequency noise. The complete MSLKAD framework, which synergizes these modules, achieves the best overall performance, confirming that balancing global context modeling with local detail refinement is essential for effective image deraining.

Table 4: Quantitative evaluations on model efficiency. Memory and latency are measured on 256×256 resolution images.

Method	Restormer [21]	IDT [11]	MSDT [5]	Ours
Params	26.10M	16.39M	14.49M	9.31M
Memory	668MB	1285MB	1096MB	758MB
Latency	85.7ms	79.2ms	75.8ms	68.4ms

To provide deeper insights into the Multi-scale Large Kernel design within the MLA, we evaluate various kernel configurations in Table 2. The results indicate that relying solely on single-scale kernels limits the network's receptive field, leading to suboptimal performance. In contrast, the multi-scale strategy achieves the highest PSNR and SSIM. This quantitative advantage is visually corroborated by the feature activation maps in Figure 4. As observed, the branch with smaller kernels (3-5-1) exhibits weak and fragmented responses, failing to capture the full extent of the rain streaks. Conversely, the large-kernel branch (7-9-1) displays strong, continuous activations that align perfectly with the spatial trajectory of the rain. These visualizations confirm that larger kernels are indispensable for encompassing the structural integrity of rain streaks, while the multi-branch architecture effectively aggregates these diverse spatial contexts to distinguish rain artifacts from complex background textures.

4. CONCLUSION

We propose the Multi-Scale Large Kernel Attention Deraining Network (MSLKAD) to tackle spatially-varying rain streaks. Our framework integrates the MLA, GKG, and EKA modules to synergistically capture multi-scale long-range dependencies, refine local details, and maintain global consistency. This unified feature extraction mechanism efficiently handles heterogeneous rain patterns with reduced complexity, striking an optimal balance between accuracy and efficiency. Comprehensive experiments on synthetic and real-world benchmarks validate MSLKAD's state-of-the-art performance and strong generalization, highlighting its po-

tential for high-quality image restoration scenarios.

- MSLKAD achieves the best performance on **five of six benchmarks**, demonstrating strong generalization across synthetic and real-world rainy conditions.
- MSLKAD achieves **33.09 dB** on Rain200H and **50.12 dB** on SPA-Data, while consistently improving performance on RealRain-1K-L/H and LHP-Rain.
- The complete MSLKAD improves PSNR by **3.05 dB** over the baseline, while using only **9.31M** parameters with **68.4 ms** inference latency.

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